

Es. 1) Divide and conquer. $T(n) = 4T(n/2) + n^2 \quad n > 1, \quad T(1) = 1 \quad n = 1.$ (unfolding)

SVOLGIMENTO:

1. Divide and conquer: Ogni problema è suddiviso in 4 sottoproblemi di costo $\frac{n}{2}$. Il costo della ricombinazione è n^2

2. Unfolding.

$$\begin{aligned}
 T(n) &= 4T(n/2) + n^2 \rightarrow T(n/2) = 4T(n/4) + (n/2)^2 \rightarrow T(n/4) = 4T(n/8) + (n/4)^2 \rightarrow T(n/8) = 4T(n/16) + (n/8)^2 \\
 \Rightarrow T(n) &= 4 \left(4 \left(4 \left(4T(n/16) + (n/8)^2 \right) + (n/4)^2 \right) + (n/2)^2 \right) + n^2 \\
 &= 4^4 T(n/2^4) + 4^3 \left(\frac{n}{2^3} \right)^2 + 4^2 \left(\frac{n}{2^2} \right)^2 + 4^1 \left(\frac{n}{2^1} \right)^2 + 4^0 \left(\frac{n}{2^0} \right)^2 \\
 &= 4^4 T\left(\frac{n}{2^4}\right) + n^2 \sum_{i=0}^3 \left(\frac{4}{4}\right)^i = 4^4 T\left(\frac{n}{2^4}\right) + n^2 \sum_{i=0}^3 1^i
 \end{aligned}$$

In generale: $T(n) = 4^K T\left(\frac{n}{2^K}\right) + n^2 \sum_{i=0}^{K-1} 1^i = 4^K T\left(\frac{n}{2^K}\right) + n^2 \cdot (K-1+1)$

Pongo $\frac{n}{2^K} = 1 \rightarrow n = 2^K \rightarrow K = \log_2 n$, allora: $T(n) = 4^{\log_2 n} \cdot 1 + n^2 \cdot \log_2 n =$
 $= n^2 + n^2 \cdot \log_2 n = O(n^2 \cdot \log n)$

Es. 2) Catene montaggio

tempi lavorazione

0	7	4	6	5
1	2	8	3	4
	0	1	2	3

tempi trasferimento

0	1	2	1
1	2	1	3
	0	1	2

tempi entrata

	2	4
0	0	1

tempi uscita

	2	1
0	0	1

SVOLGIMENTO:

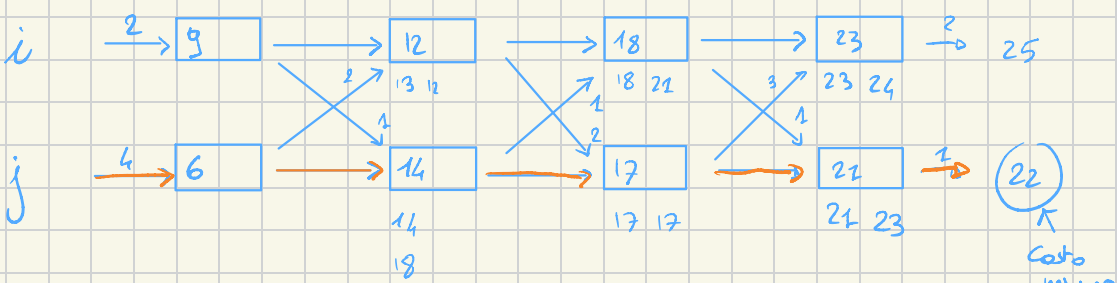


Tabella costi f:

0	9	12	18	23
1	6	14	17	21

Tabella passaggi l:

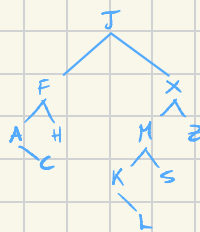
0	1	0	0
1	1	1	1

Es. 3) Infixa $\rightarrow$ Prefissa: $(2 \times 3 + 4/5) \times 8 - 4$

SVOLGIMENTO:

Prefissa: "prima l'operatore" $- \times + \times 2 3 / 4 5 8 4$

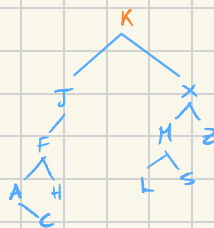
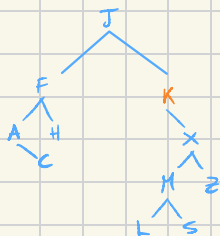
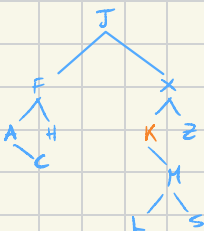
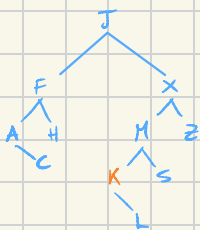
Es. 4) BST partitioning attorno a chiave mediana



SVOLGIMENTO:

Visita in-order: 1 2 3 4 5 6 7 8 9 10 11
 A C F H J K L M S X Z

$\uparrow$
 K è la chiave mediana



Es. 5) Tabella hash, $N=7$: 138 18 103 60 213 28 98 Open addressing double hashing, $\alpha < 0.4$

SVOLGIMENTO:

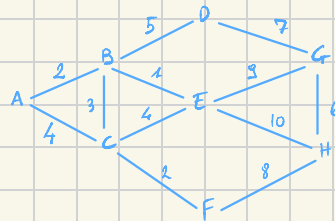
$$\alpha = \frac{N}{M} \Rightarrow M = \frac{N}{\alpha} = \frac{7}{0.4} = 17.5 \Rightarrow M=19 \text{ (numero primo)}$$

$$h(k) = (K \times 19 + i(1 + K \times 9)) \times 19$$

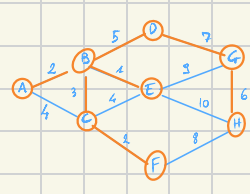
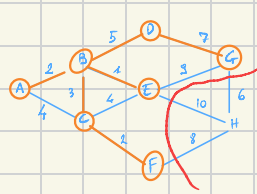
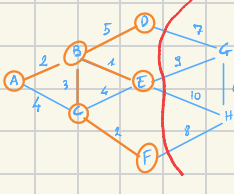
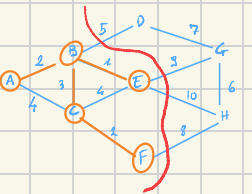
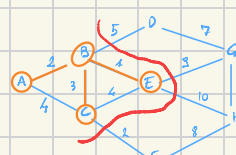
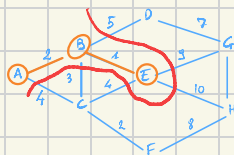
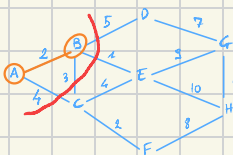
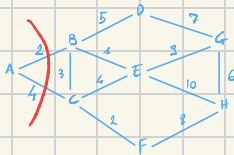
0		8	103	16	
1		9	28	17	
2		10	18	18	
3	60 (33 bit!)	11			
4	213	12			
5	138	13			
6		14			
7	98	15			

- $98 \times 19 = 3 \text{ bit!}$
- $(3 + 1(1 + 1)) \times 19 = 5 \text{ bit!}$
- $(3 + 2(2)) \times 19 = 7$

Es. 6) MST con Prim del vertice A.



SVOLGIMENTO:



$$\text{Tot: } 2 + 3 + 1 + 5 + 2 + 2 + 6 = 26$$